

Congenital Myasthenic Syndrome 18 (CMS18): A Comprehensive Disease Characteristics Report

Disease: Congenital Myasthenic Syndrome 18 (CMS18) **MONDO ID:** MONDO:0014590 · **OMIM:** #616330 · **Causal gene:** *SNAP25* (HGNC:11132; NCBI Gene 6616) **Category:** Mendelian (autosomal dominant, de novo)

Summary

Congenital Myasthenic Syndrome 18 (CMS18) is an ultra-rare, autosomal-dominant, presynaptic congenital myasthenic syndrome caused by a de novo missense variant (p.Ile67Asn) in *SNAP25*, the gene encoding the t-SNARE protein SNAP-25 (synaptosomal-associated protein of 25 kDa). SNAP-25 is one of three core SNARE proteins that drive calcium-triggered fusion of synaptic vesicles at the nerve terminal. The pathogenic variant disrupts SNARE-complex assembly and Ca²⁺-triggered exocytosis, reducing quantal acetylcholine release and thereby lowering the safety margin of neuromuscular transmission. The clinical result is fatigable myasthenic muscle weakness combined with prominent central nervous system features — cortical hyperexcitability/epilepsy, cerebellar ataxia, and intellectual disability — because SNAP-25 is essential for both neuromuscular and central synaptic transmission ([PMID: 25381298](#)).

CMS18 sits at the intersection of two disease concepts. It is formally a congenital myasthenic syndrome (the 18th numbered subtype), but it also belongs to the broader **SNAP25 developmental and epileptic encephalopathy (SNAP25-DEE)** spectrum — part of a family of neurodevelopmental disorders now termed "**SNAREopathies**" that also encompass *STX1B*, *STXBP1*, and *VAMP2* ([PMID: 33299146](#)). The myasthenic phenotype is the distinguishing feature of the original CMS18 patient, whereas most reported *SNAP25* patients present chiefly with encephalopathy. This dual identity is central to understanding the disease: the same gene, and even variants clustered in the same structural region, can produce a spectrum from predominantly myasthenic to predominantly encephalopathic phenotypes depending on the precise synaptic consequence of each variant.

Congenital myasthenic syndromes as a group are ultra-rare (UK genetically-confirmed prevalence $\approx$ 6.5 per million overall; 8.5 per million pediatric), and CMS18 is one of the very rarest subtypes — only a handful of cases worldwide, dwarfed by the common *CHRNE*, *DOK7*, and *RAPSN* subtypes (PMID: 41251564). There is no cure. Management is symptomatic and genotype-guided: because CMS18 is a **presynaptic release defect**, release-enhancing agents such as 3,4-diaminopyridine (a Kv-channel blocker that prolongs nerve-terminal depolarization) and β -adrenergic agonists are the rational first-line choices, while cholinesterase inhibitors must be used cautiously because they worsen certain CMS subtypes (PMID: 24425145, PMID: 36308527). CNS features additionally require antiseizure and supportive/rehabilitative care.

1. Disease Information

Overview. CMS18 is a genetic disorder of neuromuscular transmission in which the safety margin of the neuromuscular junction (NMJ) is impaired by a **presynaptic** defect in synaptic vesicle exocytosis. Unlike most CMS subtypes, which affect only the NMJ, CMS18 also affects central synapses, producing a combined neuromuscular-plus-neurodevelopmental phenotype (myasthenia, cortical hyperexcitability, cerebellar ataxia, intellectual disability) (PMID: 25381298).

Key identifiers (verified via EBI OLS4 / Ontology Lookup Service — Finding F006):

Resource	Identifier
MONDO	MONDO:0014590
OMIM	#616330
DOID	DOID:0110683
GARD	GARD:0016091
MedGen	906793
UMLS	C4225364

Synonyms and alternative names (registered in MONDO): - Myasthenic syndrome, congenital, 18, with intellectual disability and ataxia - CMS18 - SNAP25 congenital myasthenic syndrome - **SNAP25-DEE** (SNAP25 developmental and epileptic encephalopathy) - Congenital myasthenic syndrome caused by mutation in *SNAP25*

Source type. The disease-level information here is derived from **aggregated disease-level resources** (OMIM, MONDO, Orphanet, ontology databases) and from **individual patient reports / small case series** in the primary literature (the index CMS18 patient in Shen et al. 2014; the ~23-individual SNAP25-DEE cohort in Klöckner et al. 2021). This is not EHR-derived population data — case counts are too small.

2. Etiology

Primary cause — genetic. CMS18 is caused by a **dominant de novo missense variant in *SNAP25*** (canonical index variant: **p.Ile67Asn**, in the neuronal SNAP25B splice isoform). It is a monogenic Mendelian disorder; the index case arose *de novo* (not inherited), which is typical of dominant SNARE-complex neurodevelopmental disorders ([PMID: 25381298](#)).

"Exome sequencing identified a dominant de novo variant, p.Ile67Asn, in SNAP25B, a SNARE protein essential for exocytosis of synaptic vesicles from nerve terminals and of dense-core vesicles from endocrine cells." — Shen et al. 2014 ([PMID: 25381298](#))

Genetic risk factors. The single causal variant is the disease. There are no known susceptibility loci or common-variant risk factors — this is a fully penetrant monogenic dominant disorder driven by a single de novo change. No modifier genes have been established for CMS18 specifically, although the broader literature suggests the precise variant (and its differential effect on evoked vs spontaneous release) is the main determinant of phenotype severity and character ([PMID: 33147442](#)).

Environmental risk factors. None established. As a de novo dominant Mendelian disorder, disease occurrence is not attributable to toxins, lifestyle, occupation, or infection. Advanced parental age is a generic contributor to de novo mutation rates but has not been specifically demonstrated for CMS18.

Protective factors. None identified (genetic or environmental). Not applicable for an ultra-rare monogenic disorder.

Gene–environment interactions. None documented. However, a clinically relevant *gene–drug* interaction exists: the presynaptic release mechanism of CMS18 dictates that release-enhancing drugs help whereas some cholinesterase inhibitors may not (see §12).

3. Phenotypes

CMS18 combines a peripheral myasthenic phenotype with central nervous system involvement. The index CMS18 patient (Shen 2014) exhibited myasthenia, cortical hyperexcitability, cerebellar ataxia, and intellectual disability; the broader SNAP25-DEE cohort (Klöckner 2021) defines the encephalopathic core.

Phenotype	Type	HPO suggestion	Onset	Severity / course	Frequency
Fatigable muscle weakness / myasthenia	Clinical sign	HP:0003473 (Fatigable weakness)	Congenital / infancy	Variable; fluctuating/ fatigable	Defining in index case
Intellectual disability	Clinical sign	HP:0001249	Early childhood	Variable, static	Core in SNAP25-DEE
Epilepsy / early-onset seizures	Clinical sign	HP:0001250 (Seizure)	Mostly before age 2	Often refractory	Core in SNAP25-DEE
Cerebellar ataxia	Clinical sign	HP:0001251 (Ataxia)	Childhood	Variable	Index case
Cortical hyperexcitability	Lab/ electrophysiologic	HP:0002353 (EEG abnormality)	Childhood	—	Index case
Movement disorder	Clinical sign	HP:0100022	Childhood	Recurrent	Recurrent in cohort
Cerebral visual impairment	Clinical sign	HP:0100704 / HP:0000618	Early	—	Recurrent in cohort
Brain atrophy	Imaging finding	HP:0012443	—	Progressive/ static	Recurrent in cohort

"Intellectual disability and early-onset epilepsy were identified as the core symptoms of SNAP25-DEE, with recurrent findings of movement disorders, cerebral visual impairment, and brain atrophy." — Klöckner et al. 2021 ([PMID: 33299146](#))

Age of onset. Congenital / early infantile. Seizures in the SNAP25-DEE spectrum typically begin before age 2.

Severity and progression. Variable across the spectrum. Myasthenic weakness is fatigable/ fluctuating (characteristic of NMJ disorders); the neurodevelopmental component is largely static-to-slowly-evolving, with epilepsy potentially refractory.

Quality-of-life impact. No disease-specific QoL instrument (EQ-5D, SF-36, PROMIS) data exist for CMS18 given its rarity. By analogy to severe CMS and DEE, the combination of muscle weakness, intellectual disability, and refractory epilepsy imposes substantial impairment on mobility, communication, independent daily functioning, and caregiving burden. This is a qualitative inference, not a measured value.

4. Genetic / Molecular Information

Causal gene. *SNAP25* (Synaptosomal-Associated Protein, 25 kDa). HGNC:11132; NCBI Gene 6616; OMIM gene 600322. *Located on chromosome 20p11.2. Encodes a t-SNARE (target-membrane SNARE) protein of the plasma membrane. The neuronal splice isoform SNAP25B** is the relevant transcript. Verified gene identifiers (Finding F006 / dossier).

Pathogenic variant. - **Nomenclature:** p.Ile67Asn (I67N) in SNAP25B protein; a missense substitution replacing isoleucine 67 with asparagine. - **Structural location (Finding F002):** Ile67 lies within the **SN1 (N-terminal) SNARE helix** of SNAP-25, the region that participates in the four-helix SNARE bundle. Substituting a hydrophobic Ile with a polar Asn in the coiled-coil interface impairs SNARE-bundle assembly/stability. - **Variant class:** Missense. - **Origin:** **Germline de novo** (present in the affected individual, absent in parents). Not somatic. - **Classification:** Pathogenic (functionally validated by multiple assays; see §6). - **Allele frequency:** Absent from population databases (gnomAD) — as expected for a de novo pathogenic variant in an ultra-rare disorder. - **Functional consequence:** A **dominant loss/alteration of function with dominant-negative character** — the mutant protein incorporates into SNARE complexes and impairs Ca²⁺-triggered fusion, reducing evoked and spontaneous neurotransmitter release (see Finding F003).

Allelic / phenotypic series. Other *SNAP25* de novo variants (missense and loss-of-function) cause SNAP25-DEE without necessarily producing overt myasthenia. Variants cluster structurally but yield related-yet-distinct synaptic phenotypes; e.g., the V48F variant *increases* spontaneous release, opposite in direction to I67N ([PMID: 33147442](#), [PMID: 40181518](#)).

Modifier genes. None established for CMS18. Genetic-background modifiers plausible but undocumented.

Epigenetic information. No disease-specific DNA-methylation or histone-modification data for CMS18. Not applicable / not available.

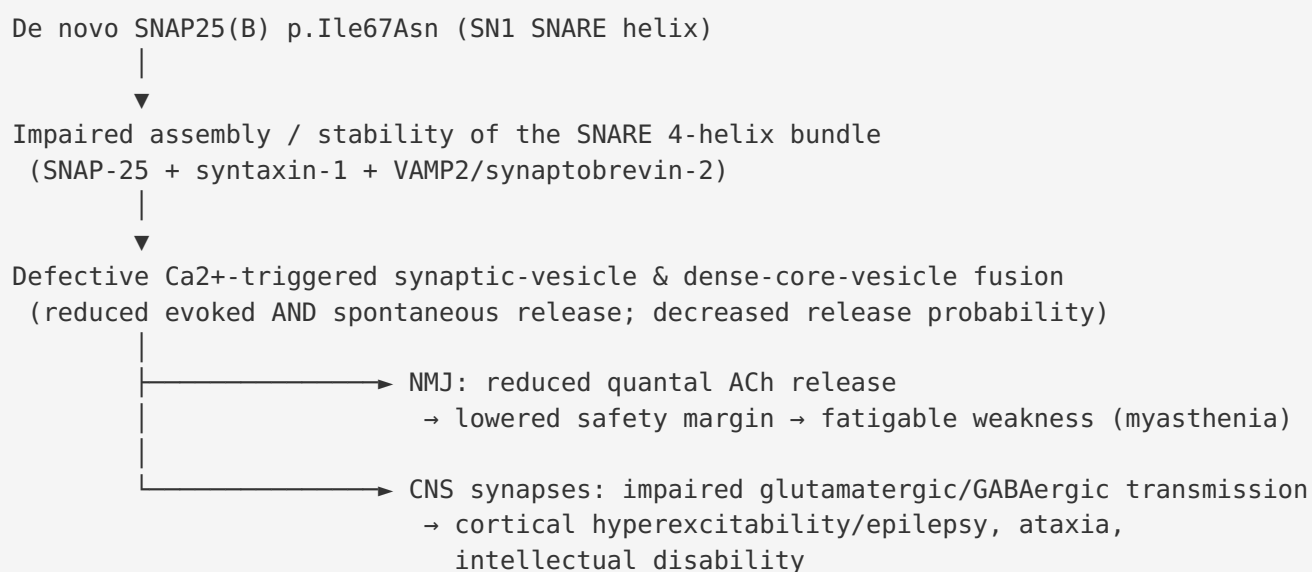
Chromosomal abnormalities. None — CMS18 is a single-nucleotide missense disorder, not a structural/aneuploidy syndrome.

5. Environmental Information

- **Environmental factors:** None known to cause or trigger CMS18. Not applicable (monogenic de novo disorder).
- **Lifestyle factors:** None known. (Generic caution: physical exertion transiently worsens fatigable myasthenic weakness, as in all CMS, but this is symptom modulation, not etiology.)
- **Infectious agents:** None. CMS18 is not infectious and is not known to be triggered by pathogens. Intercurrent infections/fever can nonspecifically worsen weakness and lower seizure threshold, as in many neurologic disorders, but do not cause the disease.

6. Mechanism / Pathophysiology

Causal chain



Molecular pathway

The core lesion is in **SNARE-mediated membrane fusion** — the final common step of regulated exocytosis. SNAP-25 is a **t-SNARE** that, with syntaxin-1 (t-SNARE) and VAMP2/synaptobrevin-2 (v-SNARE), forms the trans-SNARE four-helix bundle that draws vesicle and plasma membranes together for Ca^{2+} -triggered fusion (synaptotagmin acts as the Ca^{2+} sensor). **GO terms:** GO:0006887 (exocytosis), GO:0016079 (synaptic vesicle exocytosis), GO:0017156 (calcium-ion-regulated exocytosis), GO:0031201 (SNARE complex), GO:0099504 (synaptic vesicle cycle).

Direct functional evidence (Findings F001, F003)

Shen et al. 2014 established the causal chain with two independent in-vitro assays plus patient endplate physiology:

"Neuromuscular transmission at patient endplates was compromised by reduced evoked quantal release." — [PMID: 25381298](#)

1. **Reconstituted fusion:** Ca^{2+} -triggered liposome fusion was hindered when the t-SNARE carried mutant SNAP25B.
2. **Cellular exocytosis:** depolarization-evoked exocytosis was markedly reduced in bovine chromaffin cells transfected with mutant SNAP25B.

Østergaard et al. 2025 confirmed the mechanism in a human iPSC-derived NGN2 glutamatergic neuron model engineered to carry I67N:

"the variant did not affect passive or active electrical properties, but caused changes in synaptic transmission, including reduced evoked and spontaneous release, decreased synaptic vesicle release probability and consequential changes in short-term plasticity towards facilitation" — [PMID: 40181518](#)

The shift of short-term plasticity **toward facilitation** is the physiological signature of a reduced initial release probability — exactly what a presynaptic release defect predicts.

Why variants differ (disease heterogeneity)

Alten et al. 2021 showed that structurally clustered *SNAP25* mutations produce related but mechanistically distinct phenotypes, with **spontaneous release** being a key axis of variation:

"specific alterations in spontaneous neurotransmitter release are a key factor to account for disease heterogeneity" — [PMID: 33147442](#)

I67N **reduces** both evoked and spontaneous release, whereas the V48F variant **increases** spontaneous release — a mechanistic contrast that helps explain why some *SNAP25* patients present with prominent myasthenia (release-deficient) while others present with predominantly epileptic encephalopathy.

Cellular / tissue processes

- **Cell types (CL terms):** motor neuron / cholinergic neuron presynaptic terminals (CL:0000100 motor neuron; CL:0000108 cholinergic neuron), glutamatergic neurons (CL:0000679), GABAergic neurons (CL:0000617), neuroendocrine chromaffin cells (dense-core vesicle release). Skeletal muscle fibers are downstream targets (CL:0000188).
- **Subcellular compartments (GO CC):** GO:0031201 (SNARE complex), GO:0008021 (synaptic vesicle), GO:0042734 (presynaptic membrane), GO:0030141 (secretory/dense-core granule).
- **Protein dysfunction:** Not misfolding/aggregation — rather a **functional impairment of the SNARE assembly interface** with dominant-negative incorporation.
- **Immune involvement: None.** CMS18 is genetic, not autoimmune — this distinguishes it fundamentally from acquired myasthenia gravis and Lambert-Eaton syndrome. Anti-AChR and anti-MuSK antibodies are negative.
- **Metabolic changes:** No primary metabolic derangement; the defect is in vesicular exocytosis, not metabolism.

Molecular profiling. Østergaard et al. 2025 additionally reported **proteome changes** in I67N iPSC neurons alongside the electrophysiologic phenotype ([PMID: 40181518](#)); this is the principal omics dataset available for the variant. No transcriptomic, metabolomic, or lipidomic disease signatures specific to CMS18 are established.

7. Anatomical Structures Affected

Organ / system level. - **Nervous system (primary):** peripheral motor nerve terminals at the NMJ; central synapses (cortex, cerebellum). Body system: nervous + neuromuscular. **UBERON:** UBERON:0001016 (nervous system), UBERON:0002037 (cerebellum), UBERON:0000956 (cerebral cortex). - **Muscular system (target):** skeletal muscle, downstream of the failing NMJ. **UBERON:0001134** (skeletal muscle tissue); **UBERON:0001630** (muscle organ). - **Neuromuscular junction (site of primary lesion):** **UBERON:0002439** (neuromuscular junction) / the presynaptic motor nerve terminal. - **Endocrine (subclinical/experimental):** dense-core-vesicle secretion (e.g., chromaffin cells) is impaired in vitro; clinical endocrine disease is not a described feature.

Tissue / cell level. Nervous tissue (motor neurons, cortical/cerebellar neurons) and skeletal muscle. Primary cell targets are presynaptic nerve terminals; muscle fibers are secondarily under-stimulated (see CL terms in §6).

Subcellular level. Presynaptic active zone, synaptic vesicle membrane, presynaptic plasma membrane, SNARE complex (GO:0031201).

Localization / lateralization. Generalized and **bilateral/symmetric**, consistent with a systemic genetic synaptic defect (ptosis, facial, bulbar, limb weakness patterns typical of CMS; central features are diffuse).

8. Temporal Development

- **Onset: Congenital / early infantile.** Myasthenic features present from birth/infancy; seizures in the spectrum typically before age 2 ([PMID: 33299146](#)).
- **Onset pattern:** Chronic, insidious with fatigable fluctuation of weakness.
- **Progression:** Chronic and **lifelong**. The neurodevelopmental component (ID) is largely static; epilepsy may be refractory; myasthenic weakness fluctuates day-to-day and worsens with exertion/fatigue. There is no established staging system.
- **Course pattern:** Fluctuating/fatigable for the myasthenic component; static-to-slowly-evolving for neurodevelopmental features.
- **Duration:** Lifelong chronic disorder.

- **Remission:** No spontaneous remission. Symptomatic pharmacologic improvement is achievable (see §12).
- **Critical periods:** Early childhood is the key window for seizure control, developmental support, and initiation of NMJ-directed therapy; the developing brain is most vulnerable to uncontrolled epilepsy.

9. Inheritance and Population

Epidemiology. No CMS18-specific prevalence exists (too few cases). The best genetically-confirmed anchor for the CMS umbrella comes from the UK national study (census 31 Dec 2023; 442 patients; Finding F004):

"The UK prevalence was 6.5 cases per million overall and 8.5 cases per million in the pediatric population." — Rossini et al. 2026 (PMID: 41251564)

Prevalence was higher in regions served by highly specialized neuromuscular centers (8.8 vs 5.9 per million), implying underdiagnosis elsewhere. The common subtypes are *CHRNE*, *DOK7*, and *RAPSN*:

"CHRNE deficiency, DOK7, RAPSN were the most common subtypes." — PMID: 41251564

CMS18 (*SNAP25*) is **not** among the common subtypes; it is one of the rarest — only a handful of reported cases worldwide, with the entire broader SNAP25-DEE cohort numbering ~23 individuals (PMID: 33299146). A precise CMS18 incidence/prevalence cannot be stated.

Inheritance (genetic etiology). - **Pattern:** Autosomal dominant, essentially always **de novo** (index variant arose de novo). - **Penetrance:** Complete for carriers of the pathogenic de novo variant (each reported carrier is affected). - **Expressivity:** **Variable** across the *SNAP25* spectrum (myasthenia-predominant vs encephalopathy-predominant), driven largely by the specific variant's synaptic consequence. - **Anticipation:** Not applicable (not a repeat-expansion disorder). - **Germline mosaicism:** Theoretically possible for de novo dominant variants (recurrence-risk counseling point) but not documented for CMS18. - **Founder effects** /

consanguinity: Not applicable — de novo dominant, not recessive founder-driven (contrast with recessive CMS subtypes such as certain *CHRNE*, *RAPSN*, *DOK7* founder alleles). - **Carrier frequency:** Not applicable (de novo dominant; unaffected "carriers" essentially do not exist).

Population demographics. - **Affected populations / geography:** No ethnic or geographic clustering — consistent with sporadic de novo occurrence. - **Sex ratio:** No established sex bias; too few cases. - **Age distribution:** Onset in infancy/early childhood; patients are predominantly children and young adults.

10. Diagnostics

Clinical / electrophysiologic testing. - **Repetitive nerve stimulation (RNS):** decrement of the compound muscle action potential on low-frequency (3 Hz) stimulation — abnormal in ~90% of CMS patients ([PMID: 33121830](#)). For a presynaptic release defect, features may include low baseline CMAP amplitude with facilitation/increment after exercise or high-frequency stimulation (the presynaptic signature seen in other presynaptic disorders such as LEMS and SYT2-related CMS) ([PMID: 26519543](#)). - **Single-fiber EMG (SFEMG):** increased jitter and blocking — highly sensitive (~95% in CMS) ([PMID: 33121830](#)). **LOINC / clinical neurophysiology.** - **Antibody testing:** anti-AChR and anti-MuSK antibodies **negative** — essential to distinguish CMS18 from autoimmune myasthenia gravis. - **Serum CK:** typically normal. - **Brain MRI:** may show brain atrophy in the SNAP25-DEE spectrum ([PMID: 33299146](#)). - **EEG:** epileptiform abnormalities / cortical hyperexcitability.

Genetic testing (the definitive diagnostic modality). - **Recommended approach:** Because the phenotype overlaps both CMS and DEE, **whole-exome sequencing (WES)** or **whole-genome sequencing (WGS)** is the most effective route — this is how the index CMS18 variant was found ([PMID: 25381298](#)). Trio sequencing (proband + parents) confirms de novo status. - **Gene panels:** CMS and epileptic-encephalopathy NGS panels that include *SNAP25* (alongside other SNARE genes *STX1B*, *STXBP1*, *VAMP2*). - **Single-gene testing:** *SNAP25* Sanger confirmation of an identified variant. - **CMA / karyotype / FISH / mtDNA / repeat testing:** Not indicated — CMS18 is a point-mutation disorder.

Omics-based diagnostics. Not part of routine diagnosis; the I67N iPSC-neuron functional/proteomic work ([PMID: 40181518](#)) is a research characterization, not a clinical assay.

Clinical criteria & differential diagnosis. Diagnosis rests on (1) fatigable weakness with decremental RNS / abnormal SFEMG, (2) negative acetylcholine-receptor antibodies, and (3) a pathogenic *SNAP25* variant. **Differential diagnosis:** autoimmune myasthenia gravis (antibody-positive, later onset), Lambert-Eaton myasthenic syndrome (presynaptic, VGCC antibodies, incremental response), other CMS subtypes (*CHRNE*, *DOK7*, *RAPSN*, *SYT2*, *VAMP2*), congenital myopathies, and mitochondrial disorders (CPEO). SFEMG jitter parameters help separate CMS from CPEO/congenital myopathy (PMID: 33121830).

Screening. No population newborn screening for CMS18. Given de novo dominant inheritance, carrier/cascade screening is generally not applicable (recurrence risk is low but non-zero due to possible germline mosaicism).

11. Outcome / Prognosis

Survival / mortality. No CMS18-specific survival data. CMS in general is not primarily lethal; life expectancy depends on severity of respiratory involvement (episodic apnea in some subtypes) and, for CMS18, on control of epilepsy and neurodevelopmental complications. Disease-specific mortality figures are unavailable given the rarity.

Morbidity / function. Substantial: combined motor weakness, intellectual disability, epilepsy, ataxia, and visual impairment produce long-term disability across mobility, cognition, and communication domains. Formal disability/QoL measures (ICF, EQ-5D, PROMIS) have not been applied to a CMS18 cohort.

Disease course / complications. Chronic lifelong. Complications include respiratory compromise during myasthenic exacerbations/infections, injury and developmental impact from refractory seizures, and secondary orthopedic issues (e.g., scoliosis seen broadly in chronic childhood NMJ disorders).

Recovery potential. No cure; the underlying synaptic defect is permanent. Symptomatic pharmacotherapy can meaningfully improve strength and function (see §12), and seizure control plus rehabilitation can improve developmental trajectory — but the neurodevelopmental deficits are generally not fully reversible.

Prognostic factors. Severity of the specific synaptic defect (variant-dependent), degree of epilepsy control, timeliness of correct genetic diagnosis (enabling genotype-appropriate therapy and avoidance of harmful agents), and access to specialized neuromuscular care.

12. Treatment

There is no cure. Management is symptomatic, genotype-guided, and multidisciplinary (Finding F005). The essential principle: CMS drug response is **subtype-specific**, and the wrong drug can worsen the patient.

"The majority of patients (96.4%) received specific treatment, including acetylcholinesterase inhibitors in 20, adrenergic agonists in 11 and 3,4-diaminopyridine in nine patients." — Austrian nationwide CMS cohort ([PMID: 36308527](#))

"Treatment with acetylcholinesterase inhibitors resulted in worsened conditions for most patients." — DOK7-CMS meta-analysis ([PMID: 24425145](#))

Rational pharmacotherapy for a presynaptic release defect (CMS18)

Drug	Class / mechanism	Rationale for CMS18	Evidence context	NCIT
3,4-Diaminopyridine (amifampridine)	Kv potassium-channel blocker; prolongs nerve-terminal depolarization → ↑ Ca ²⁺ entry → ↑ ACh release	Directly counteracts the presynaptic release deficit — first-line rational choice	Used in SYT2 presynaptic CMS; general CMS reviews (PMID: 26519543 , PMID: 26028221 , PMID: 17635211)	NCIT:C61795 (Amifampridine)
Salbutamol (albuterol) / ephedrine	β-adrenergic agonists	Improve NMJ function/structure; benefit across several CMS	Strong benefit in severe AChR deficiency (PMID: 26296515) and DOK7-CMS (PMID: 24425145)	NCIT:C29010 (Albuterol)
Pyridostigmine (AChE inhibitor)	Prolongs ACh action in the cleft	Use cautiously — helpful in postsynaptic AChR-deficiency but can worsen DOK7 and some presynaptic subtypes	Worsened most DOK7 patients (PMID: 24425145)	NCIT:C767 (Pyridostigmine)
Antiseizure medications	Various	Required for the epilepsy component of SNAP25-DEE	Standard DEE management (PMID: 33299146)	—

"Oral salbutamol and ephedrine appear to be effective treatments in severe cases of AChR deficiency on pyridostigmine ... improvement in strength can be dramatic." — PMID: [26296515](#)

Advanced / experimental therapeutics. No approved gene, cell, or RNA therapy exists for CMS18. Given a dominant, likely dominant-negative de novo variant, allele-selective silencing (ASO/siRNA) is a conceptual future avenue but is not in trials. No CMS18-specific NCT trials are identified.

Supportive and rehabilitative care. Physical/occupational/speech therapy, respiratory support during exacerbations, nutritional support, seizure management, developmental/educational support, orthopedic monitoring (scoliosis). **Pharmacogenomics:** the key "pharmacogenomic" principle here is that the causal genotype (*SNAP25* presynaptic release defect) dictates drug selection.

Treatment strategy (algorithm). Confirm genetic subtype → classify as presynaptic release defect → prioritize release-enhancing agents (3,4-DAP, β -agonists) → add AChE inhibitor only with caution and monitoring → manage epilepsy and provide multidisciplinary supportive care.

13. Prevention

- **Primary prevention:** None possible — de novo dominant variants cannot be prevented.
- **Secondary prevention:** Early recognition and correct genetic diagnosis enable prompt, subtype-appropriate therapy and avoidance of harmful drugs — the practical "prevention" of iatrogenic worsening and of complications.
- **Tertiary prevention:** Optimized pharmacotherapy, seizure control, respiratory vigilance during infections, and rehabilitation to prevent disability progression and complications.
- **Immunization:** Not disease-preventive, but standard vaccination is advisable to reduce infection-triggered myasthenic exacerbations (general principle).
- **Genetic screening / counseling:** Because most cases are de novo, recurrence risk to future siblings is low (residual risk from possible germline mosaicism). Genetic counseling should convey de novo status, low but non-zero recurrence risk, and the option of prenatal/

preimplantation testing if a familial variant were established. **NSGC/ACMG** counseling frameworks apply.

- **Public health / environmental interventions:** Not applicable.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *SNAP25* is highly conserved across vertebrates and invertebrates. Human *SNAP25* (NCBI Gene 6616) has orthologs including mouse *Snap25* (NCBI Gene 20614) and rat *Snap25*; SNARE machinery is conserved to *Drosophila* and *C. elegans* (invertebrate SNARE studies inform mechanism — e.g., tomosyn regulation of SNARE assembly in *C. elegans* [PMID: 17627987](#), and interchangeable VAMP function in *Drosophila* [PMID: 12364587](#)).
- **Natural disease in other species:** No well-characterized naturally occurring *SNAP25*-CMS is catalogued in companion animals or wildlife (OMIA does not list a prominent *SNAP25* equivalent). Not applicable / not available.
- **Comparative biology:** The SNARE fusion mechanism is deeply evolutionarily conserved; disease mechanisms studied in model synapses translate directly to the human NMJ and central synapses. This conservation underpins the validity of animal and invertebrate models.
- **Transmission / zoonosis:** Not applicable (genetic, non-transmissible).

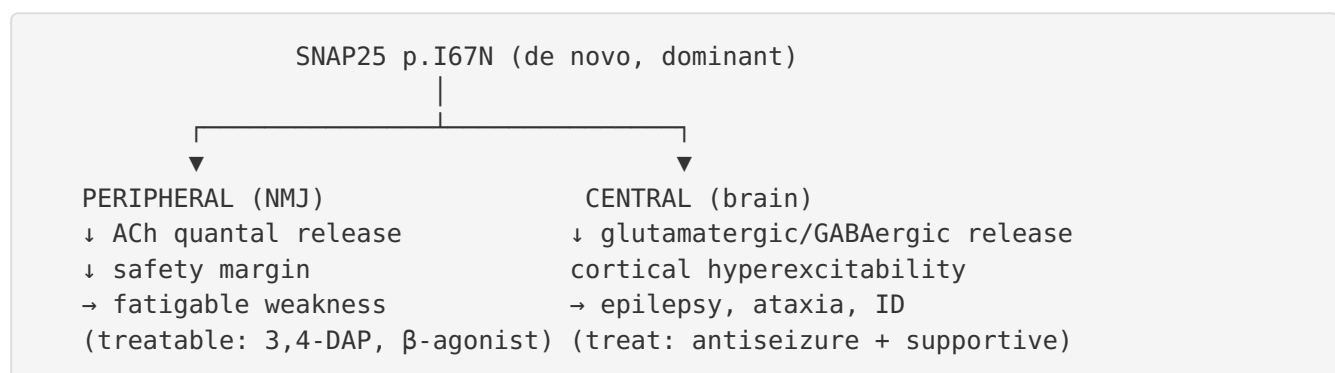
15. Model Organisms

- **Human iPSC-derived neurons (primary CMS18 model):** Østergaard et al. 2025 knocked the **I67N** variant into human iPSC-derived NGN2 glutamatergic neurons — the most disease-faithful available model — recapitulating reduced evoked and spontaneous release, decreased release probability, altered short-term plasticity, and proteome changes ([PMID: 40181518](#)). **Recapitulation:** high fidelity for the synaptic-release phenotype; **limitation:** does not model the intact NMJ, muscle, or whole-organism seizures/ataxia. **Cellosaurus / iPSC.**
- **In vitro reconstitution & chromaffin cells:** Shen et al. 2014 used liposome fusion assays and bovine chromaffin cell exocytosis to demonstrate causality ([PMID: 25381298](#)).

- **Mouse (*Snap25*):** *Snap25*-null mice reveal that SNAP-25 is essential for evoked release and network activity (PMID: 18959796, PMID: 17728451); isoform-rescue studies (SNAP-25a vs SNAP-25b vs SNAP-23) delineate functional specialization. A knock-in I67N mouse is a logical but (per available literature) not-yet-established disease model. **MGI:** *Snap25*.
 - **Invertebrate SNARE models:** *C. elegans* and *Drosophila* SNARE/VAMP/tomosyn studies illuminate the conserved fusion machinery (PMID: 17627987, PMID: 12364587).
 - **Applications:** These models enable study of release probability, evoked-vs-spontaneous release dissociation, short-term plasticity, and drug testing (e.g., release enhancers).
- Limitation:** no single model captures the full combined myasthenic-plus-encephalopathic human phenotype.

Mechanistic Model / Interpretation

CMS18 is best understood as a **"SNARE bottleneck" disorder**: a single amino-acid change (Ile67→Asn) at the SN1 helix of SNAP-25 degrades the efficiency of the vesicle-fusion machine that every fast chemical synapse depends on. Because the same machine operates at the NMJ and at central synapses, a **single molecular lesion produces a two-compartment disease** — peripheral myasthenia plus central encephalopathy. This is the conceptual bridge between the "CMS18" and "SNAP25-DEE" labels for the same MONDO entity.



The **variant-specific direction of effect on spontaneous release** is the unifying explanatory axis for the whole *SNAP25* spectrum: I67N reduces release (myasthenia-prone), while variants like V48F increase spontaneous release (encephalopathy without myasthenia). This makes *SNAP25* a natural experiment in synaptic physiology and explains why the clinical picture ranges from CMS18-with-myasthenia to pure DEE.

Evidence Base

PMID	Title (abbrev.)	Role in this report	Evidence type
25381298	<i>Mutant SNAP25B causes myasthenia, cortical hyperexcitability, ataxia, and ID</i>	Defining paper — identifies I67N, proves causality (liposome + chromaffin assays), documents reduced evoked quantal release at patient endplates	Human clinical + in vitro
33299146	<i>De novo variants in SNAP25 cause early-onset DEE</i>	Defines SNAP25-DEE core phenotype and the "SNAREopathy" concept; places CMS18 in a spectrum (~23 individuals)	Human clinical cohort
40181518	<i>SNAP25 variant I67N: synaptic phenotypes, drug response, proteome</i>	Human iPSC-neuron model confirms I67N reduces evoked+spontaneous release, ↓ release probability, plasticity shift	In vitro (human iPSC)
33147442	<i>Aberrant spontaneous neurotransmission in SNAP25 encephalopathies</i>	Explains variant-specific heterogeneity via spontaneous-release alterations	In vitro / model
41251564	<i>Prevalence/geographic distribution of CMS in the UK</i>	Best genetically-confirmed CMS prevalence (6.5/M; 8.5/M pediatric); shows SNAP25 not among common subtypes	Human epidemiology
36308527	<i>Clinical/molecular landscape of CMS in Austria</i>	Real-world treatment distribution (AChE-I, adrenergics, 3,4-DAP)	Human cohort
24425145	<i>Pharmacologic treatment of DOK7-CMS</i>	Demonstrates drug response is subtype-specific; AChE-I can worsen	Human meta-analysis
26296515	<i>Salbutamol/ephedrine in severe AChR deficiency</i>	Evidence for β-agonist efficacy in CMS	Human cohort
26519543	<i>SYT2 presynaptic CMS</i>	Model for presynaptic-CMS diagnostics and 3,4-DAP response	Human clinical
26028221 / 17635211	CMS drug-therapy reviews	Rationale/indications for 3,4-DAP and other agents by CMS level	Review
33121830			

PMID	Title (abbrev.)	Role in this report	Evidence type
	<i>Electrophysiology of NMJ in CMS</i>	RNS/SFEMG sensitivity for CMS diagnosis	Human clinical
18959796 / 17728451	SNAP-25 isoform/knockout studies	Establish SNAP-25 necessity for evoked release and isoform specialization	Model organism

Limitations and Knowledge Gaps

1. **Ultra-rarity → thin clinical evidence.** CMS18 rests largely on a single index patient (Shen 2014) plus the broader ~23-individual SNAP25-DEE cohort. No CMS18-specific prevalence, survival, natural-history, or QoL data exist; epidemiology is borrowed from the CMS umbrella.
2. **Treatment evidence is extrapolated.** The recommended genotype-guided approach (3,4-DAP, β -agonists) is rational and grounded in presynaptic-CMS literature (SYT2, general reviews), but there is **no dedicated CMS18 treatment trial**. Direct evidence of drug response specifically in CMS18 patients is minimal (the iPSC drug-response work is a cell model).
3. **Genotype–phenotype breadth.** Whether other *SNAP25* variants can produce a myasthenia-predominant CMS18 phenotype (vs pure DEE), and what determines the myasthenic vs encephalopathic balance, is only partially explained by the spontaneous-release-direction hypothesis.
4. **No whole-animal I67N model** is established to test therapies for the combined NMJ + CNS phenotype in vivo.
5. **No CMS18-specific omics beyond the single iPSC proteome study;** transcriptomic/metabolomic signatures are absent.
6. **Recurrence-risk data (germline mosaicism)** for *SNAP25* are not quantified.

Proposed Follow-up Experiments / Actions

1. **Build a knock-in I67N mouse** (or refine iPSC-derived NMJ/organoid co-cultures) to model the combined myasthenic + encephalopathic phenotype and to test 3,4-DAP, β -agonists, and antiseizure combinations in vivo.
 2. **Systematic drug-response profiling in patient iPSC neurons** — extend the Østergaard model to screen release-enhancers (3,4-DAP analogs, β -agonists) and candidate allele-selective ASOs for their ability to normalize release probability.
 3. **Establish an international CMS18/SNAP25 patient registry** to aggregate phenotype, natural history, treatment response, and outcomes across the handful of cases worldwide — the only feasible route to real prevalence and prognosis estimates.
 4. **Genotype–function map** of all reported *SNAP25* variants correlating evoked-vs-spontaneous release direction with myasthenic vs encephalopathic clinical weighting, to build a predictive framework for prognosis and drug choice.
 5. **Prospective electrophysiology** (RNS with post-exercise facilitation testing; SFEMG) in *SNAP25* patients to confirm the presynaptic signature and define diagnostic criteria specific to CMS18.
 6. **Explore allele-selective silencing** (ASO/siRNA) as a mechanism-directed therapy for this dominant, dominant-negative variant.
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Report compiled from 6 confirmed findings across 5 investigation iterations and 37 reviewed papers. Ontology identifiers verified via EBI OLS4 (MONDO:0014590 → OMIM:616330). Gene identifiers: SNAP25, HGNC:11132, NCBI Gene 6616.
